

MedGemma-27B vs Bio-Medical-Llama-3-8B — the same EEG reports, two labelings

We labeled Harvard’s clinical EEG reports — **101,614 reports across three hospitals** — with **MedGemma-27B** (prompt v5g, Q4) and compared the result against Harvard’s own **Bio-Medical-Llama-3-8B** labels for the same reports. There is no human ground truth here, so this is a **similarity comparison between two independent language models**, not an accuracy one. It is a stress test of generalization: does a model built and tuned on our own reports line up with a different model, from a different group, on reports from a hospital system it has never seen?

We compare the three findings that map cleanly between the two schemas:

MedGemma	Bio-Medical-Llama
overall Abnormality	abnormal
Focal non-epileptiform	foc slowing
Generalized non-epileptiform	gen slowing

We report **two numbers per finding: raw present/absent agreement and Cohen’s κ** (agreement beyond chance). Kappa matters because raw agreement is inflated for the rare findings, where both models say the finding is absent most of the time, so a high raw number can hide a low real similarity. BIDMC (Beth Israel) is left out: its report text is not in the shared data.

Overall — how much the labels coincide

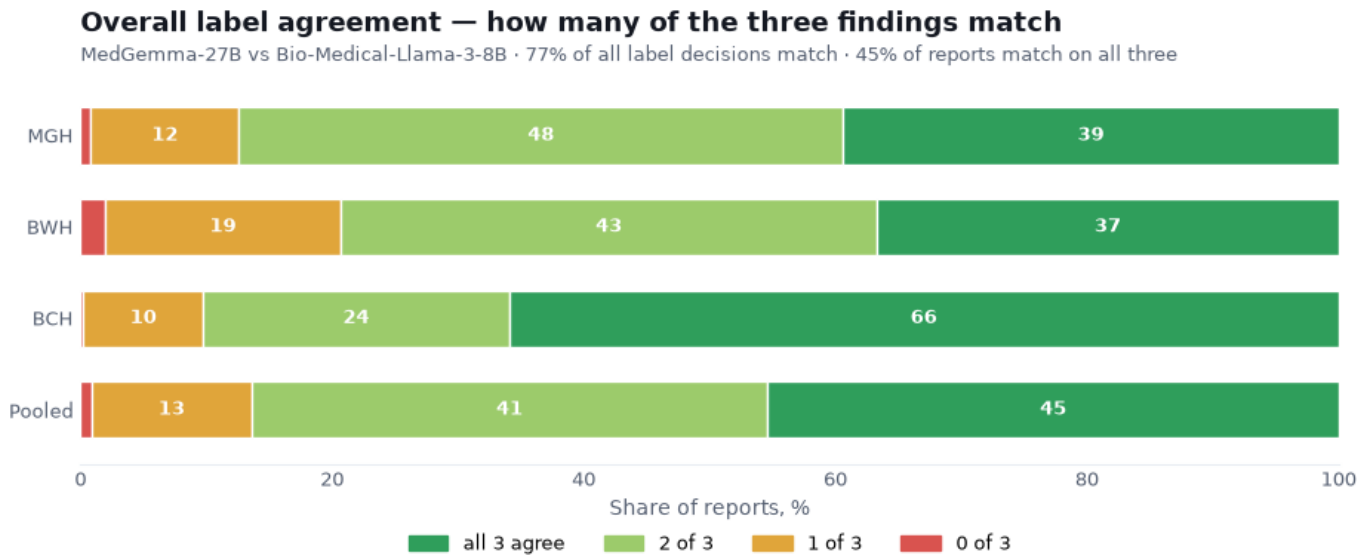


Figure 1: Overall agreement

Across all findings, **77% of individual label decisions match** between the two models. Counting whole reports, **45% agree on all three findings**, another 41% on two of three, and only ~1% on none. The pediatric hospital (BCH) is the closest match.

Per finding — how similar the two really are

Finding	MGH agree / κ	BWH agree / κ	BCH agree / κ	Pooled agree / κ
Abnormality	91% / 0.80	83% / 0.58	98% / 0.96	91% / 0.81
Focal slowing	72% / 0.35	65% / 0.35	78% / 0.27	72% / 0.35

Finding	MGH agree / κ	BWH agree / κ	BCH agree / κ	Pooled agree / κ
Gen. slowing	62% / 0.26	66% / 0.28	80% / 0.53	67% / 0.36

The **overall abnormal-or-normal call is strongly shared — $\kappa = 0.81$** (and 0.96 at BCH). On a hospital system neither model was built for, the two independently agree on whether an EEG is abnormal, well beyond chance.

The **slowing findings look agreed on the raw number (67–72%), but κ is only 0.35–0.36** — fair agreement. Most of that raw 70% is just both models agreeing the finding is absent; on the reports that actually mention slowing, the two often disagree.

By hospital — match rate and κ

Two views side by side. **Match rate** counts every report where the two models make the same present/absent call; κ discounts the easy both-absent agreement and shows the real similarity.

On the raw match rate everything looks 62–98% similar. But κ separates it: abnormality is a real match ($\kappa = 0.96$ at BCH, 0.80 at MGH, 0.58 at BWH — at BWH the two call abnormality at somewhat different rates), while the slowing findings are only fair everywhere ($\kappa = 0.26$ –0.53) — most of their 62–80% match rate is just both models agreeing that the finding is absent.

Where they diverge — the call rate

How often each model marks the finding present (pooled):

Finding	MedGemma-27B (Q4)	Bio-Medical-Llama-3-8B
Abnormality	67%	64%
Focal slowing	40%	15%
Gen. slowing	46%	64%

On overall abnormality the two call it at nearly the same rate (67% vs 64%). The slowing findings split, and in **opposite directions**: MedGemma marks **focal slowing about three times as often** (40% vs 15%), while Bio-Medical-Llama marks **generalized slowing more** (64% vs 46%). The two models simply divide slowing between focal and generalized differently.

What this means

- **The main result is the abnormality agreement ($\kappa = 0.81$).** Two independently built models, on an unseen hospital’s reports, agree on the overall call well beyond chance — good evidence that the labeling generalizes past the reports it was tuned on.
- **The slowing divergence is most likely definitional, not a plain error.** With no human label, neither model is definitively correct here, and the two disagree in *opposite* directions on focal vs generalized, which points to the two schemas drawing the focal/generalized line for slowing differently. Bio-Medical-Llama’s labels are also part rule-based and curated, marked present mainly when a finding is stated explicitly, whereas MedGemma infers from the narrative — which would push it to call more where the wording is suggestive.

Caveats

- **Similarity, not accuracy.** No human ground truth for these reports, so where the two differ neither can be called correct.
- Only the three cleanly-mappable findings are compared. Epileptiform is split differently on the two sides (Bio-Medical-Llama does not separate spikes/seizures into focal vs generalized) and is left for a separate look.

How similar the two labelings are, per finding

Raw agreement is inflated for the rare findings; κ shows the real similarity · pooled MGH+BWH+BCH

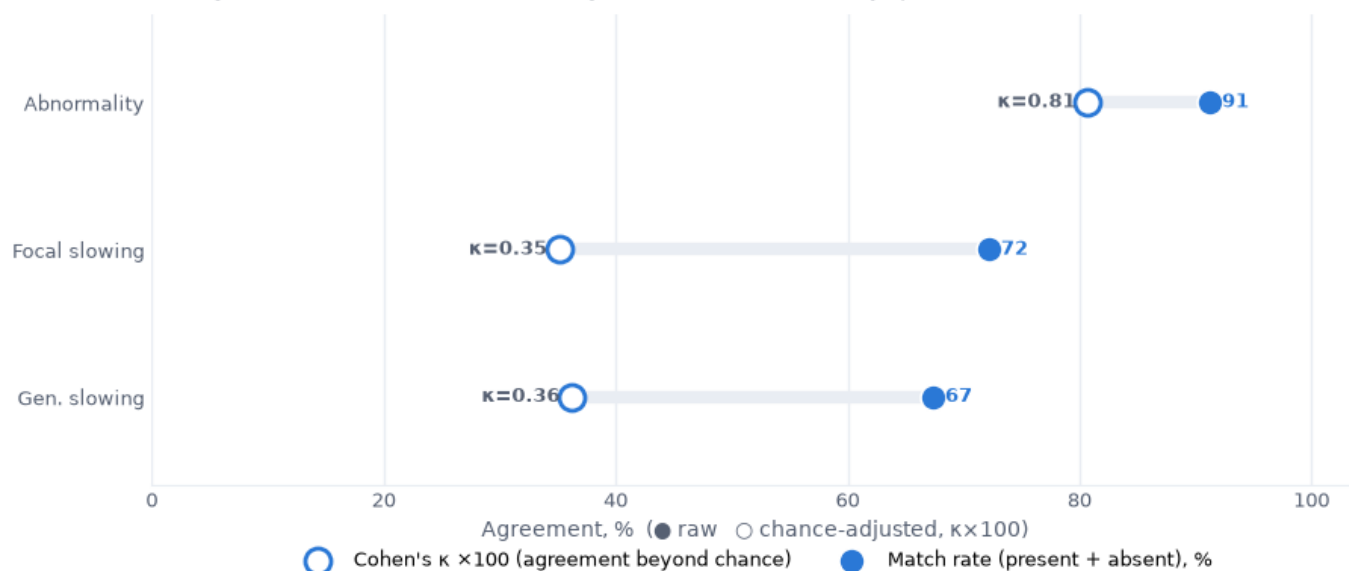


Figure 2: Agreement and kappa

How often the two labelings match, by hospital

Present/absent match rate per finding · MedGemma-27B vs Bio-Medical-Llama-3-8B



Figure 3: Match rate by hospital

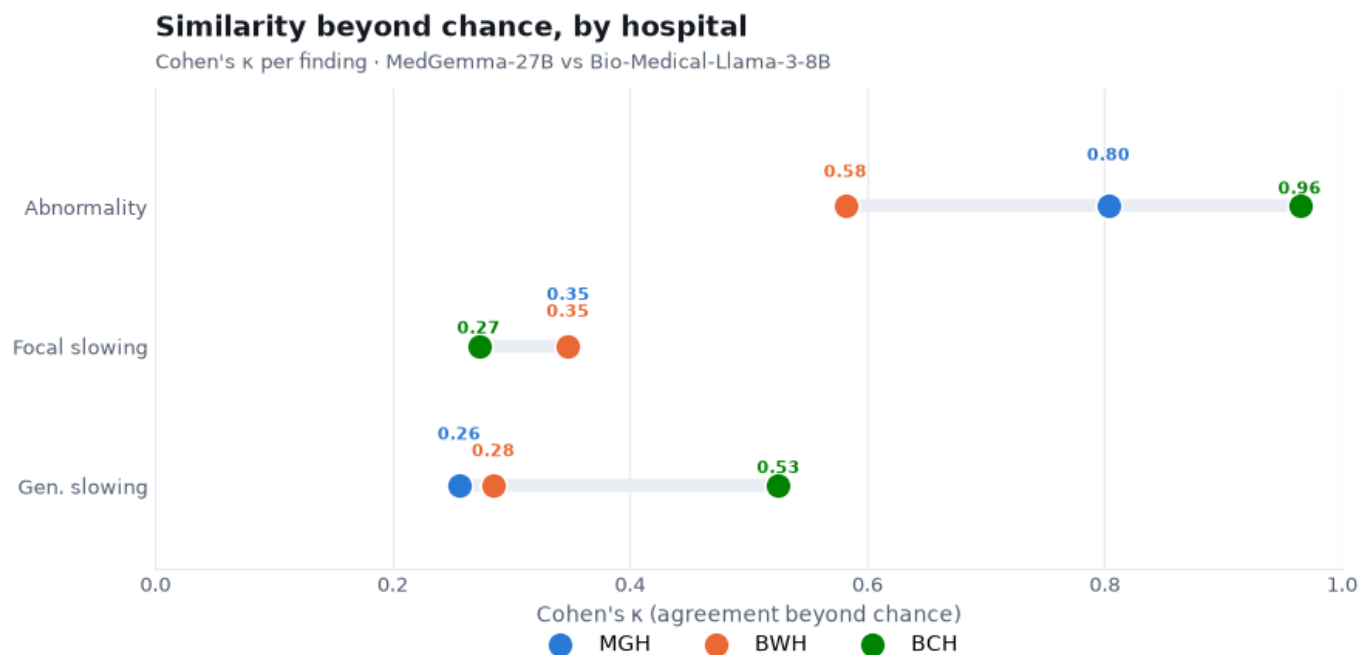


Figure 4: Kappa by hospital

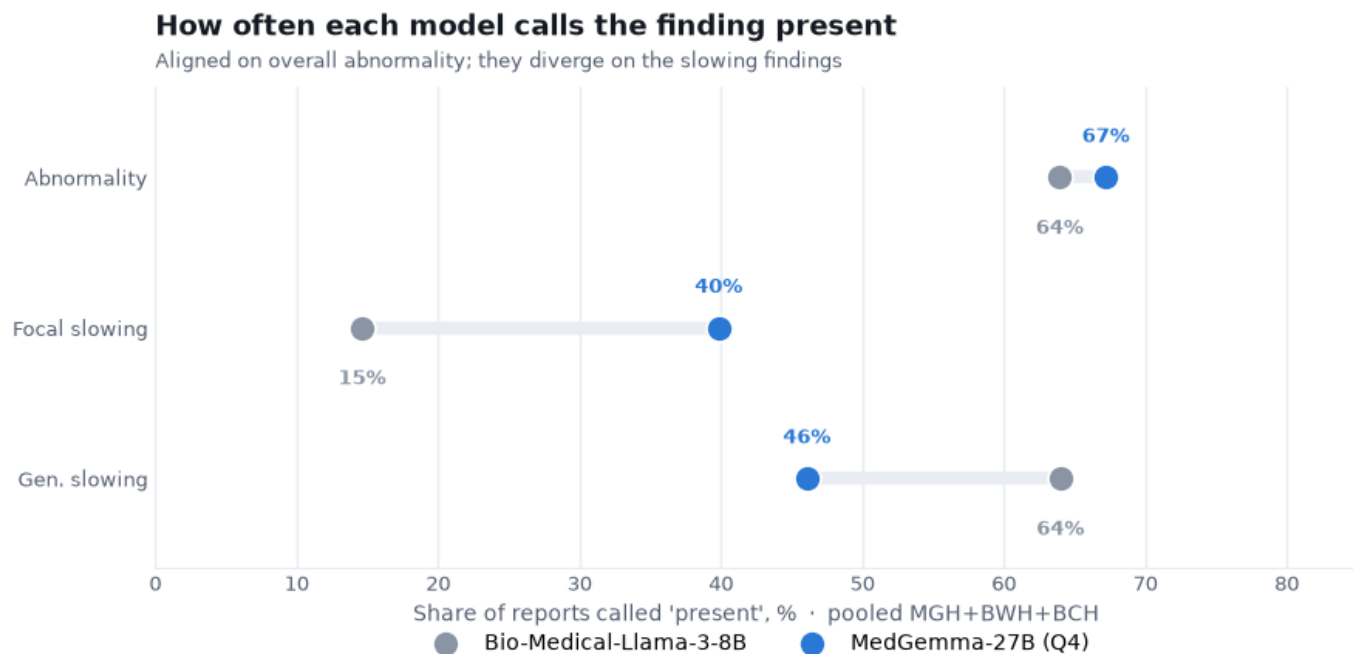


Figure 5: Call rate

- **BIDMC excluded** — report text unavailable.

Reproduce

```
python -m analysis.harvard_compare      # the tables above + all four figures
```

MedGemma labels: results/harvard/labels/*/ (produced by slurm/label_harvard.sbatch via slurm/submit_harvard.sh, reading the report text from the HEEDB zips through analysis/harvard_index.py). Bio-Medical-Llama labels: the *_EEG_with_reports tables in HarvardEEG.db.

Pipeline, prompts, and the scripts behind this report: **<https://github.com/vbronetskyi/eeg-report-classification>**